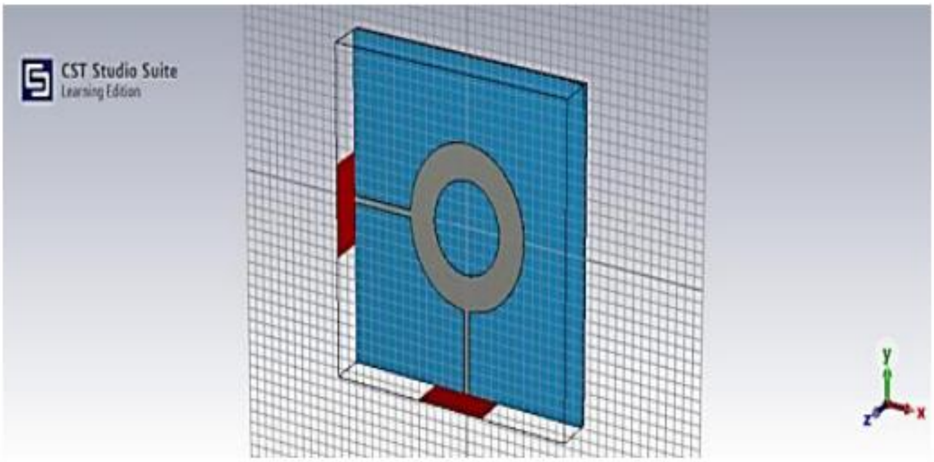
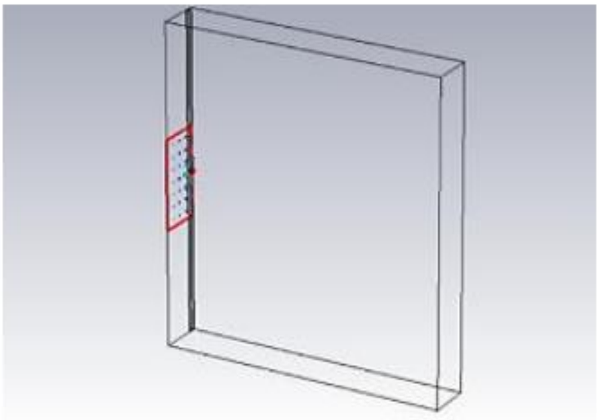
EXNO: 14 Electromagnetic Field and Power Distribution Analysis of a Circular Resonator in CST Studio Suite.

Test Case 1: Analyze the electric field and magnetic field distribution at Port 1of the circular resonator and examine the electric and magnetic field parameters at Port 1 for studying electromagnetic field behavior and validating the system performance.

Simulation Results:



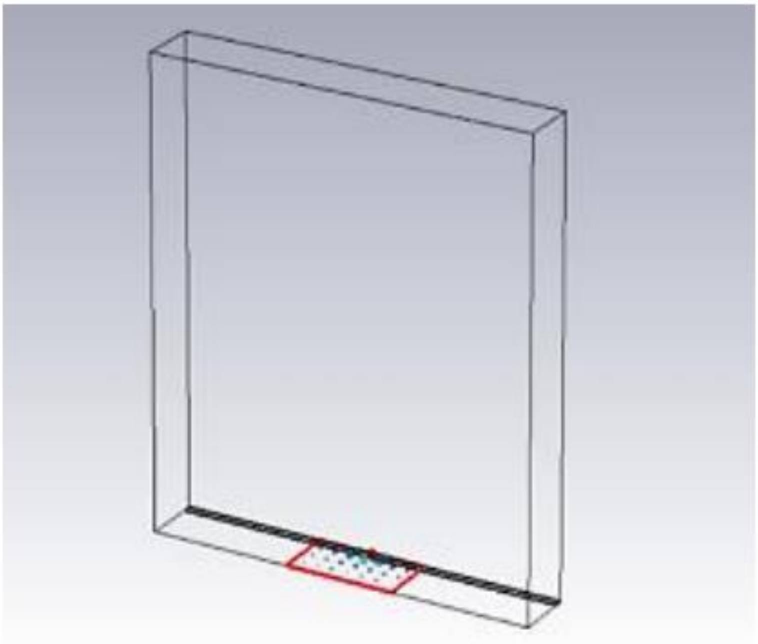


Output:

Parameter	Value
e1 (Electric Field)	10796.2 V/m
h1 (Magnetic Field)	74.274 A/m
Frequency	5 GHz
Mode Type	Quasi TEM
Wave Impedance	47.7744 Ω
Line Impedance	141.945 Ω
Phase Constant β	278.911 1/m
Peak Value	10796.2 V/m

Test Case 2: Analyze the electric field and magnetic field distribution at Port 2of the circular resonator and examine the electric and magnetic field parameters at Port 2 for studying electromagnetic field behavior and validating the system performance.

Simulation Results:



Output:

Parameter	Value
e2 (Electric Field)	10796.2 V/m
h2 (Magnetic Field)	76.274 A/m
Frequency	5 GHz
Mode Type	Quasi TEM
Wave Impedance	141.945 Ω
Line Impedance	47.7744 Ω
Phase Constant β	278.911 1/m
Peak Value	e2 = 10796.2 V/m, h2 = 76.274 A/m